PHYSICS

1) A stone is dropped into a quiet lake and waves move in circles spreading out radially at the speed of 5 cm/s. At the instant when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing ?

(1) 80 π cm^2/s
(2) 90 π cm^2/s
(3) 85 π cm^2/s
(4) 40 π cm^2/s

2) The position of a particle moving along the y-axis is given as $y = 3t^2 - t^3$ where y is in metres and t is in sec. The time when the particle attains maximum positive y position will be :

(1) 1.5 sec
(2) 4 sec
(3) 2 sec
(4) 3 sec

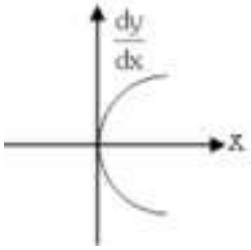
3) sin75° =

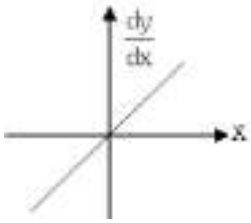
(1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
(2) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
(3) $\frac{\sqrt{2}+1}{2}$
(4) None

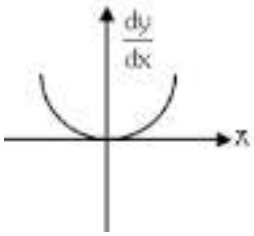
4) $\int (3\cos x - 4\sin x)dx =$

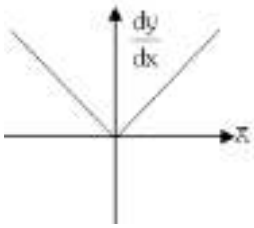
(1) –3 sinx – 4 cosx + C
(2) +3 sinx – 4 cosx + C
(3) 3 sinx + 4 cosx + C
(4) –3 sinx + 4 cosx + C

5) If $y = \frac{x^3}{3}$ then, the graph between $\frac{dy}{dx}$ & x-axis is:-

(1) 

(2) 

(3) 

(4) 

6) The sum of the series $1-\frac{1}{3}+\frac{1}{9}-\frac{1}{27}+\frac{1}{81}\ldots\infty$

(1) $\frac{3}{4}$

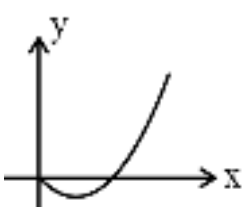
(2) $\frac{4}{3}$

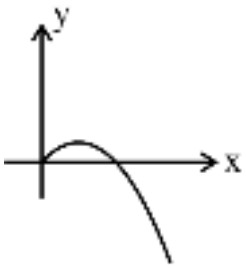
(3) $\frac{4}{6}$

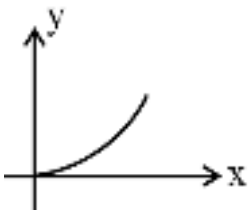
(4) $\frac{4}{9}$

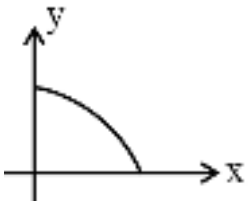
7)

Which of following curve represent $y = 2x - x^2$?

(1) 

(2) 

(3) 

(4) 

8) If $y = t^2$ and $x = 5t$ than find $\frac{dy}{dx}$

(1) 0.5

(2) $\frac{2.5}{t}$

(3) 2t

(4) 0.4t

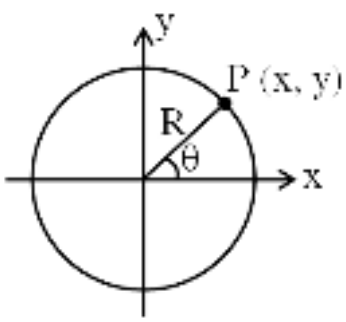
9) $\int_{3e}^{4e} \frac{dx}{x+e} =$

(1) $\log\left(\frac{5}{4}\right)$

(2) $\ell n\left(\frac{5}{4}\right)$

(3) $\frac{5}{4}$

(4) 1

10) 

Co-ordinates of particle P are:

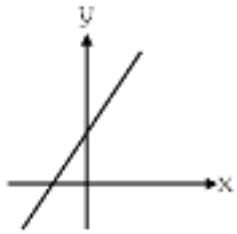
(1) (R cosθ, R sinθ)

(2) (R cosθ, – R sinθ)

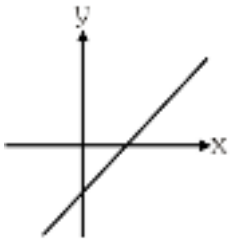
(3) (R sinθ, R cosθ)

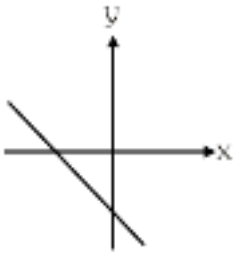
(4) (–R sinθ, R cosθ)

11) Graph of equation 5x – 2y + 4 = 0 is :-

(1) 

(2) 

(3) 

(4) 

12) The maximum and minimum value of (5 – 3 sin θ) respectively are:

(1) 8 and 0

(2) 2 and 0

(3) 8 and 2

(4) 5 and 2

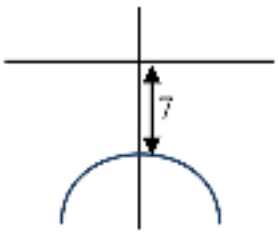
13) Differentiation of $\cos\left(2\sqrt{x}\right)$ with respect to x is :

(1) $\frac{\sin(2\sqrt{x})}{\sqrt{x}}$

(2) $-\frac{\sin(2\sqrt{x})}{\sqrt{x}}$

(3) $\frac{\sin(2\sqrt{x})}{2\sqrt{x}}$

(4) $-\frac{\sin(2\sqrt{x})}{2\sqrt{x}}$



14) From the figure the equation of parabola can be :-

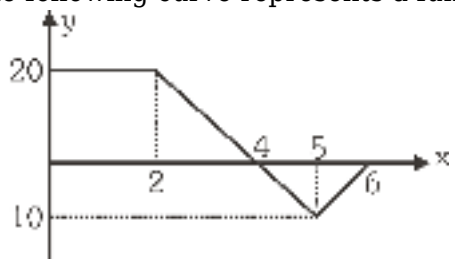
(1) $(y - 7) = x^2$

(2) $y = x^2 + 7$

(3) $y = -(x + 7)^2$

(4) $(y + 7) = -x^2$

15) The following curve represents a function y = f(x). Find average value of y when x changes from 0 to 6.



(1) $\frac{20}{3}$

(2) $\frac{25}{3}$

(3) $\frac{35}{3}$

(4) $\frac{40}{3}$

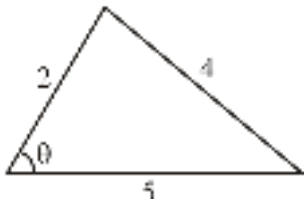
16) A wheel makes 360 revolution in 60 minute. Through how many radian does it turn in one second?

(1) $\frac{\pi}{3}$rad

(2) $\frac{\pi}{5}$rad

(3) $\frac{\pi}{4}$rad

(4) $\frac{\pi}{6}$rad

17) In ΔABC, the value of cosθ is : 

(1) $\frac{15}{16}$

(2) $\frac{13}{20}$

(3) $\frac{17}{20}$

(4) $\frac{3}{16}$

18) Value of $\cos\left(\frac{11\pi}{6}\right)$:-

(1) $\frac{-\sqrt{3}}{2}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

19) Find sum of given arithmetic progression :
{4 + 7 + 10 + ... + 94}

(1) 31

(2) 30

(3) 1519

(4) 1470

20) A particle moves along the curve $12y = x^3$. Which coordinate changes at faster rate at x = 10 ?

(1) x-coordinate

(2) y-coordinate

(3) Both x and y coordinate

(4) Data insufficient

21) The value of $\log_{10}(32)$ is :- ($\log_{10}2 = 0.3010$)

(1) 1.505

(2) 1.405

(3) 2.4

(4) 0.8

22)

If $\sin\theta = \frac{1}{3}$ then $\tan\theta$ = ?

(1) $\frac{1}{2\sqrt{2}}$

(2) $2\sqrt{2}$

(3) $\frac{2\sqrt{2}}{3}$

(4) $\frac{1}{3}$

23) **Statement-1 :** $\frac{1}{\sqrt{1-\frac{t}{T}}}$ can be written as $1+\frac{t}{2T}$ when t is very small with respect to T.
Statement-2 : $(1+x)^n = 1+nx$

(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

(4) Statement-1 is false, Statement-2 is false

24) A particle is moving with speed 6 m/s along the direction of $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$, then its velocity is :

(1) $(4\hat{i} + 2\hat{j} - 4\hat{k})$ units

(2) $(4\hat{i} + 4\hat{j} - 2\hat{k})$ units

(3) $(4\hat{i} + 4\hat{j} - 4\hat{k})$ units

(4) $(2\hat{i} + 4\hat{j} - 2\hat{k})$ units

25) Which of the following is a unit vector ?

(1) $\hat{i} + \hat{j}$

(2) $\cos\theta\hat{i} - \sin\theta\hat{j}$

(3) $\sin\theta\hat{i} + 2\cos\theta\hat{j}$

(4) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j})$

26) If direction of $\vec{P}$ is reversed then, $\left|\vec{\Delta P}\right|$ = ...

(1) P

(2) 2P

(3) 0

(4) $-2\vec{P}$

27) The unit vector parallel to the resultant of the vectors $\vec{A} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 2\hat{i} + 4\vec{j} - \hat{k}$:-

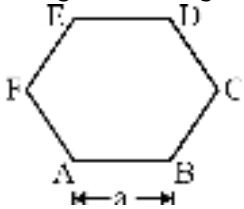
(1) $\frac{1}{49}\left(7\hat{i} + 6\hat{j} - 2\hat{k}\right)$

(2) $\frac{1}{7}\left(3\hat{i} + 6\hat{j} - 2\hat{k}\right)$

(3) $\frac{1}{49}\left(3\hat{i} + 6\hat{j} - 2\hat{k}\right)$

(4) $\frac{1}{7}\left(7\hat{i} + 6\hat{j} - 2\hat{k}\right)$

28) A regular hexagon of side 'a' shown in figure. If $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = K\overrightarrow{AD}$, the value of K is :-



(1) 3

(2) 4

(3) 5

(4) 6

29) If $\vec{A}$ and $\vec{B}$ are two non–zero vectors such that $|\vec{A} + \vec{B}| = \frac{|\vec{A} - \vec{B}|}{2}$ and $|\vec{A}| = 2|\vec{B}|$ then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 37°

(2) 53°

(3) $\cos^{-1}(-3/4)$

(4) $\cos^{-1}(-4/3)$

30) For two vectors $\vec{a}$ and $\vec{b}$, if $\vec{R} = \vec{a} + \vec{b}$ and $\vec{S} = \vec{a} - \vec{b}$, If $2|\vec{R}| = |\vec{S}|$ when $\vec{R}$ is perpendicular to $\vec{a}$, then –

(1) $\frac{a}{b} = \sqrt{\frac{3}{7}}$

(2) $\frac{a}{b} = \sqrt{\frac{7}{3}}$

(3) $\frac{a}{b} = \sqrt{\frac{1}{5}}$

(4) $\frac{a}{b} = \sqrt{\frac{5}{1}}$

31) If $\vec{a}_1$ and $\vec{a}_2$ are two non collinear unit vectors and $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then the value of $(\vec{a}_1 - \vec{a}_2)$.

$(2\vec{a}_1 + \vec{a}_2)$ is –

(1) 2

(2) $\frac{3}{2}$

(3) $\frac{1}{2}$

(4) 1

32) Given that $\vec{A} + \vec{B} = \vec{R}$ and $\vec{A} + 2\vec{B}$ is perpendicular to $\vec{A}$, then -

(1) 2 | B | = | R |

(2) | B | = 2 | R |

(3) | B | = | R |

(4) $B^2 = 2R^2$

33) The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{P}$ and $\vec{Q}$ is 2 : 1. Then $|\vec{P}|$ is equal to :

(1) $4|\vec{Q}|$

(2) $3|\vec{Q}|$

(3) $2|\vec{Q}|$

(4) $|\vec{Q}|$

34) Which of following is incorrect:-

(1) All equal vectors are parallel vectors

(2) All opposite vector are antiparallel vectors

(3) Two vectors are always coplanar vectors

(4) All vectors are coplanar vectors

35) The component of $\vec{A} = \hat{i} + \hat{j} + 5\hat{k}$ perpendicular to $\vec{B} = 3\hat{i} + 4\hat{j}$ is

(1) $-\frac{4}{25}\hat{i} + \frac{3}{25}\hat{j} + 5\hat{k}$

(2) $-\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$

(3) $\frac{4}{25}\hat{i} - \frac{3}{25}\hat{j} + 5\hat{k}$

(4) $+\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$

36) The greatest and the least resultant of two forces acting at a point is 10N and 6N, respectively. If each forces is increased by 3N, find the resultant of new forces when acting at a point at an angle of 90° with each other :-

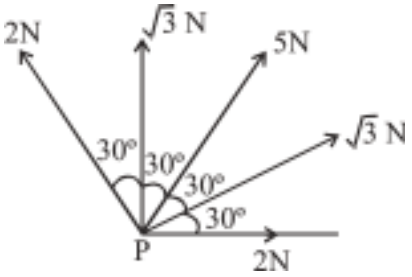
(1) $\sqrt{146}\,N$

(2) $\sqrt{124}\,N$

(3) $\sqrt{188}\,N$

(4) $\sqrt{114}\,N$

37) Five forces 2 N, $\sqrt{3}$ N, 5 N, $\sqrt{3}$ N and 2 N respectively act at a particle P as shown in the figure.



The resultant force on the particle P is

(1) 10 N making angle 60º with X-axis

(2) 10 N making angle 60º with Y-axis

(3) 20 N along Y-axis

(4) None of these

38)

Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ bear the following relations. $\vec{a}-\vec{b}+\vec{c}=\vec{0}$, $b=\sqrt{a^2+c^2}$ and $a=\sqrt{3}c$.

Angle between vectors.....

(1) $\vec{a}$ and $\vec{b}$ is 30°

(2) $\vec{a}$ and $\vec{b}$ is 60°

(3) $\vec{b}$ and $\vec{c}$ is 30°

(4) $\vec{a}$ and $\vec{c}$ is 60°

39) What is the unit vecotr perpendicular to the following vecotrs $2\hat{i}+2\hat{j}-\hat{k}$ and $6\hat{i}-3\hat{j}+2\hat{k}$?

(1) $\frac{\hat{i}+10\hat{j}-18\hat{k}}{5\sqrt{17}}$

(2) $\frac{\hat{i}-10\hat{j}+18\hat{k}}{5\sqrt{17}}$

(3) $\frac{\hat{i}-10\hat{j}-18\hat{k}}{5\sqrt{17}}$

(4) $\frac{\hat{i}+10\hat{j}+18\hat{k}}{5\sqrt{17}}$

40) Two adjacent sides of a parallelogram are represented by the two vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$. What is the area of parallelogram ?

(1) 8

(2) $8\sqrt{3}$

(3) $3\sqrt{8}$

(4) 192

41)

The component of the vector $\vec{A} = 2i + 3j$ along the vector $i + j$ is

(1) 5

(2) $\frac{5}{\sqrt{2}}$

(3) $10\sqrt{2}$

(4) $2\sqrt{5}$

42)

If vectors $\vec{A} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is :

(1) $t = 0$

(2) $t = \frac{\pi}{4\omega}$

(3) $t = \frac{\pi}{2\omega}$

(4) $t = \frac{\pi}{\omega}$

43) If $\left|\vec{A} \times \vec{B}\right| = \sqrt{3}\,\vec{A} \cdot \vec{B}$ find $\left|\vec{A} + 2\vec{B}\right|$:-

(1) $(A^2 + B^2 + AB)^{1/2}$

(2) $(A^2 + 4B^2 + 4AB)^{1/2}$

(3) $(A^2 + 4B^2 + 2AB)^{1/2}$

(4) none of these

44) Match the columns :-

	Column (I)		Column (II)
(A)	$\left(3\hat{i} + 2\hat{j} + 3\hat{k}\right) . \left(\hat{i} - \hat{j} + \hat{k}\right)$	(P)	0
(B)	$\left\|\left(\hat{i} + \hat{j} + \hat{k}\right) \times \left(2\hat{i} - \hat{j} + 2\hat{k}\right)\right\|$	(Q)	$3\sqrt{2}$
(C)	$\left(\hat{a} + \hat{b}\right) . \left(\hat{a} \times \hat{b}\right)$	(R)	4
(D)	$\left(\hat{a} + \hat{b}\right) . \left(\hat{a} - \hat{b}\right)$	(S)	2

(1) A – R, B – Q, C – P, D – S

(2) A – S, B – R, C – S, D – P

(3) A – R, B – Q, C – P, D – P

(4) A – R, B – S, C – P, D – P

45) If three vectors $x\hat{a} - 2\hat{b} + 3\hat{c}$, $-2\hat{a} + y\hat{b} - 4\hat{c}$ and $-z\hat{b} + 2\hat{c}$ are coplanar, where $\hat{a}$, $\hat{b}$ and $\hat{c}$ are unit vectors, along three perpendicular directions, then

(1) xy + 3 xz – 3z – 4 = 0

(2) xy – 2xz – 3z – 4 = 0

(3) 2xz – 3xz + 3z – 4 = 0

(4) xy – 2xz + 3 z – 4 = 0

CHEMISTRY

1) The total number of protons in 10g of calcium carbonate is :-

(1) 1.505×10^{24}

(2) 2.04×10^{24}

(3) 3.011×10^{24}

(4) 4.09×10^{24}

2) What mass of sodium is required to have same numbers of atoms that are present in 6 gm of magnesium?

(1) 23 g

(2) 46 g

(3) 5.75 g

(4) 11 g

3) Which of the following weight the most?

(1) 1g of nitrogen atom

(2) One mole of water

(3) One mole of sodium

(4) One molecule of H_2SO_4

4) Atomic weight of elements A and B are 30 and 60 respectively. If there are 'y' atoms in the x gm of A. How many atoms would be there in 2x gm of element B :-

(1) y

(2) 2y

(3) 0.5y

(4) 1.5y

5) Ribose has the formula $C_5H_{10}O_5$. What is the mass of ribose sample containing 1.25×10^{24} carbon atoms :-

(1) 23.9 g

(2) 1.99 g

(3) 150.1 g

(4) 62 g

6) An organic compound contains 4% sulphur. Its minimum molecular weight is :

(1) 200

(2) 400

(3) 800

(4) 1600

7) 4.4 g of an unknown gas occupies 2.24 L of volume of NTP. The gas may be:

(1) C_2H_6

(2) CO

(3) N_2O

(4) CH_4

8) Which of the following contains the largest mass of hydrogen atoms?

(1) 5.0 moles $C_2H_2O_4$

(2) 1.1 moles $C_3H_8O_3$

(3) 1.5 moles $C_6H_8O_6$

(4) 4.0 moles $C_2H_4O_2$

9) If mass of helium is 16 amu. How many atoms present in it?

(1) 4

(2) $4 \times N_A$

(3) N_A

(4) 2

10) An organic compound with C = 40%, H = 6.7 % and rest is oxygen will have the empirical formula :-

(1) CH

(2) CH_2O

(3) C_3H_6O

(4) $C_3H_4O_2$

11) Which of the following has the least number of molecules :-

(1) 22.4×10^3 mL of CO_2 gas at NTP

(2) 22 g of CO_2 gas

(3) 11.2 L of CO_2 gas at NTP

(4) 0.1 mole of CO_2 gas

12) If the density of gas at STP is 0.2 g/L. Then find out molecular weight of gas.

(1) 2.24
(2) 10
(3) 4.48
(4) 1

13) Mole fraction of ethanol in ethanol-water mixture is 0.25. Hence, percentage concentration of ethanol (C_2H_6O) by weight of mixture is :-

(1) 25
(2) 75
(3) 46
(4) 54

14) Volume of a drop of water is 0.0018 mL. Then number of moles in 10 drops of water (density = 1 g/mL) :-

(1) 10^{-2}
(2) 10^{-3}
(3) 10^{-4}
(4) 10^{-5}

15) Dissolving 120g of urea (molar mass = 60g) in 1000g of water gave a solution of density 1.14 g/ml. The molarity of the solution is :

(1) 1.78 M
(2) 2.00 M
(3) 2.05 M
(4) 2.22 M

16)

A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine, the formula of the metal chloride will be:-

(1) MCl_2
(2) MCl_4
(3) MCl_5
(4) MCl_3

17) A compound contains 28% nitrogen and 72% of a metal by weight. If 3 atoms of the metal combine with 2 atoms of nitrogen, then atomic weight of the metal is :-

(1) 24

(2) 28

(3) 32

(4) 42

18) **Statement–1 :** 1 mole O_3 = N molecule O_3 = 3N atom of O = 48 g.
Statement–2 : A mole is the amount of matter that contains as many as objects as the number of atoms exactly in 12 g C^{12}.

(1) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1.

(2) Statement–1 is false, Statement–2 is true.

(3) Statement–1 is true, Statement–2 is false.

(4) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

19)

The volume of 0.1 mol SO_2 gas at 27ºC and 152 mm Hg will be :-

(1) 1.23 L

(2) 12.31 L

(3) 0.123 L

(4) 123.1 L

20) A 25.0 mm × 40.0 mm piece of gold foil is 0.25 mm thick. The density of gold is 19.32 g/cm^3. How many gold atoms are in the sheet ? (Atomic weight : Au = 197.0)

(1) 7.7×10^{23}

(2) 1.5×10^{23}

(3) 4.3×10^{21}

(4) 1.47×10^{22}

21) The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is :

(1) 68.4 g

(2) 17.1 g

(3) 34.2 g

(4) 136.8 g

22) Two metallic oxides contain 27.6% and 30.0% by weight oxygen respectively. If the formula of the first oxide is M_3O_4, then that of the second will be :-

(1) MO

(2) MO_2

(3) M_2O_3

(4) M_2O_5

23) In a compound every C atom has 3 Hydrogen atoms and every 2H atom have 5 oxygen atoms. The formula of compound may be :-

(1) $C_2H_6O_{15}$

(2) $C_3H_7O_{14}$

(3) $CH_3O_{7.5}$

(4) $C_4H_8O_{16}$

24) The hydrated salt $CaCl_2.xH_2O$ observed 24.5% loss in mass on heating and becomes anhydrous. The value of x is :-

(1) 1

(2) 2

(3) 3

(4) 4

25) What value of 75% alcohol by weight (d = 0.80 g/cm^3) must be used to prepare 150 cm^3 of 30% alcohol by weight (d = 0.90 g/cm^3) :-

(1) 67.5 mL

(2) 56.25 mL

(3) 44.44 mL

(4) None of these

26) In a compound of X and Y, X is 40% by mass of compound and atomic weight of X is 60% of Y. Empirical formula is :-

(1) X_9Y_8

(2) $X_{10}Y_9$

(3) X_9Y_{10}

(4) X_8Y_9

27)

Concentrated HCl contains 38% HCl by mass. The molarity of this solution is
(density = 1.19 g/cm^2) :-

(1) 7.25 M

(2) 9.32 M

(3) 12.37 M

(4) 15.27 M

28) Two solutions labelled as 3M HCl and 1 M HCl are mixed in the ratio of x : y by volume and the molarity of resulting solution becomes 1.5M. What is the molarity of the resulting solution if they are mixed in the ratio of y : x by volume?

(1) 4 M

(2) 3 M

(3) 2 M

(4) 2.5 M

29) **Statement-1 :** When a solution is diluted from volume V_1 to V_2 by adding solvents, its molarity before dilution M_1 and after dilution M_2 are related as :
$M_1V_1 = M_2V_2$
and
Statement-2 : After dilution moles of solvent remains conserved.

(1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

30)

50 ml of '20V' H_2O_2 is mixed with 200 ml, '10V' H_2O_2. The volume strength of resulting solution is ?

(1) 10

(2) 12

(3) 16

(4) 18

31)

Molarity and % w/W of a solution of NaOH is 5 and 15. The density of solution is :-

(1) 1

(2) 1.33

(3) 1.67

(4) 2.33

32)

When 400 g of a 20% solution by weight was cooled, 50 g of solute precipitated. The percentage concentration of remaining solution is

(1) 8.57%

(2) 15%

(3) 12.25%

(4) 9.5%

33) A given mixture consists only of pure substance X and pure substance Y. The total weight of mixture is 3.72 g. The total number of moles is 0.06. If the weight of one mole of Y is 48 g and there is 0.02 mole of X in mixture. What is weight of one mole of X

(1) 40

(2) 60

(3) 80

(4) 90

34) A molecule of a compound have 13 carbon atoms, 12 Hydrogen atom, 3 oxygen atoms and 3.02×10^{-23} gm of other element, then molecular weight of compound :-

(1) 100 amu

(2) 234 amu

(3) 400 amu

(4) 234 gm

35) You are asked by your teacher to boy 10 moles of distilled water from a shop where small bottles each containing 20g of such water are available. How many bottles will you buy :-

(1) 8

(2) 9

(3) 10

(4) 11

36) The density of air is 0.001293 gm/ml. It's vapour density is :-

(1) 143

(2) 14.3

(3) 1.43

(4) 0.143

37) The average atomic mass of a mixture containing 79 % of ^{24}Mg and remaining 21 % of ^{25}Mg and ^{26}Mg is 24.31. % mol of ^{26}Mg is

(1) 5

(2) 20

(3) 10

(4) 15

38)

Vapour density of mixture of NO_2 and N_2O_4 is 34.5, then percentage abundance of NO_2 in mixture is

(1) 50%

(2) 25%

(3) 40%

(4) 60%

39) At what temperature, 2.8 g of carbon monoxide at 0.821 atm pressure will be 3 litres ? (R =

0.0821 L-atm-mol^{-1}-K^{-1})

(1) 350 K
(2) 320 K
(3) 300 K
(4) 273 K

40) Nitrogen forms several gaseous oxides. One of them has a density of 1.33 g/L measured at 764 mmHg and 150ºC. Write the formula of the compound.

(1) NO
(2) N_2O
(3) NO_2
(4) N_2O_5

41)

A gas is found to have a formula $[CO]_x$. If its vapour density is 70 the value of x is

(1) 2.5
(2) 3.0
(3) 5.0
(4) 6.0

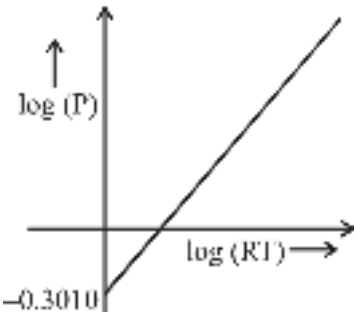
42) 0.2 of a gas X occupies a volume of 0.44 L at same pressure and temperature. Under identical conditions of P and T, 0.1 g of CO_2 gas occupies 0.32 L. Gas X can he

(1) O_2
(2) SO_2
(3) NO
(4) C_4H_{10}

43) 16 g of oxygen and 3 g of hydrogen are mixed and kept at 760 mm pressure and 0°C. The total volume occupied by the mixture will be nearly

(1) 22.4 litres
(2) 33.6 litres
(3) 448 litres
(4) 44800 ml

44) log (P) vs log RT curve is plotted for 1 mol of an ideal gas.Calculate molar volume occupied by



gas.

(1) 1 L

(2) 3 L

(3) 2 L

(4) 0.5 L

45) Density of a equimolar mixture of CH_4 and CO at STP is :-

(1) 22.0 gL^{-1}

(2) 11.2 gL^{-1}

(3) 1.96 gL^{-1}

(4) 0.98 gL^{-1}

BIOLOGY

1) Which of the following statement is incorrect ?

(1) Unicellular organisms are capable of independent existance

(2) Any structure less than a complete structure of a cell does not ensure independent living

(3) Matthias schleiden, a German zoologist, Proposed that body of animals are composed of cells and product of cells

(4) Schleiden and Schwann together formulated the cell theory

2) Cell component, that was first described by Robert Brown, is not found in which of the following cells?

(1) Ostrich egg

(2) Mesophyll cells

(3) Erythrocyte of mammals

(4) Nerve cells

3) The ribosomes of eukaryotic cytoplasm, mitochondria and plastids are respectively:

(1) 80S, 70S and 70S

(2) 70S, 70S and 80S

(3) 50S, 30S and 70S

(4) 70S, 80S and 70S

4)

German botanist who examined large number of plants and observed that all plants are composed of different kind of cells which form the tissue in the plants :-

(1) Schwann
(2) Matthias schleiden
(3) Rudolf virchow
(4) Antonvon Leeuwenhoek

5)

Read the given statement carefully
(A) All cells arise from pre existing cell
(B) Schwann is a British Botanist, examined large number of plants.
(C) All living organism are composed of cell and cell products.
(D) Phrase omnis cellula-e cellula was given by Schleiden

(1) 2 - correct, 2 - incorrect
(2) 1 - correct, 3 - incorrect
(3) 3 - correct, 1 - incorrect
(4) 4 - correct, 0 - incorrect

6)

Which of the following statement is not a part of modern cell theory?

(1) Cell has a thin outer layer called plasma membrane
(2) All living organisms are made up of cells and products of cells
(3) All cells arise from pre-existing cells
(4) All

7)

A typical eukaryotic cell consists of :-
(a) Cell membrane (b) Nucleus
(c) Cytoplasm (d) Glycocalyx

(1) a + b
(2) b + c
(3) a + b + c + d
(4) a + b + c

8) In ____,____ a German botanist examined a large number of plants and observed that all plants are composed of different kinds of cells which form tissue of the plants :- **[NCERT Pg : 88]**

(1) 1839, Leeuwenhoek
(2) 1838, Schleiden

(3) 1839, Schwann

(4) 1838, Schwann

9) **Statement A :** Ribosomes are membraneless organelles found in all cells.
Statement B : Ribosomes are found within cytoplasm, mitochondria, chloroplast and on rough ER

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both statements are incorrect

(4) Both statements are correct

10)

Nucleus was discovered by -

(1) Robert Lavies

(2) Mavesh

(3) Robert Brown

(4) Mischer

11)

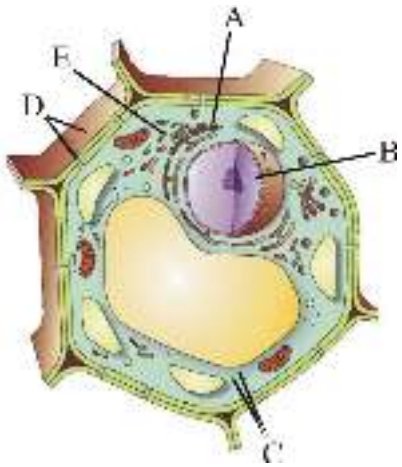
Unicellular organisms are capable of

(1) Independent existence

(2) Show the essential function of life

(3) Labour division

(4) Both 1 & 2



12)
Choose the correct option for A to E.

(1) A-RER, B-Nucleus, C-Ribosome, D-Plasmodesmata, E-SER

(2) A-SER, B-Nucleus, C-Ribosome, D-Plasmodesmata, E-RER

(3) A-SER, B-Nucleus, C-Ribosome,
D-Plasmodesmata, E-RER

(4) A-RER, B-Nucleus, C-Chloroplast,
D-Microtubules, E-SER

13)

Who proposed the hypothesis that the bodies of animals and plants are composed of cells and product of cells?

(1) Robert Hooke

(2) Schwann

(3) Both (a) and (b)

(4) None of these

14)

How many of the following are correct statement:-
(i) Rudolf virchow first explained that cell divide and new cells are formed from pre existing cells
(ii) Cell wall is the main arena of cellular activity.
(iii) The shape of the cell may vary with the function they perform.
(iv) Schleiden based on his studies concluded that the presence of cell wall is unique character of plant cells.

(1) Two

(2) Three

(3) Four

(4) One

15) Physico-chemical approach to study and understand living organism is called

(1) Comparative biology

(2) Reductionist biology

(3) Biophysics

(4) Biochemistry

16)

Who for the first time reported that the most striking difference between animal and plant is of cell wall -

(1) Schleiden

(2) Schwan

(3) Virchow

(4) Nagelli

17) Read the following statements (A–C) carefully and choose the correct option :- **[NCERT NO. 88]**
(A) Chemical reactions occur in cytoplasm keep the cell in the "living state"

(B) Rudolf virchow first explained that cells divided and new cells formed from pre-existing cells
(C) Eukaryotic cells multiply more rapidly than the prokaryotic cells

(1) Only (C) is incorrect
(2) (A) & (C) are correct
(3) (B) & (C) are correct
(4) All statements are correct

18) Match column I with column II and select the correct option from the codes given below :

	Column I		**Column II**
(A)	Leeuwenhoek	(i)	First saw and described a living cell
(B)	Robert Brown	(ii)	Presence of cell wall is unique to plant cells
(C)	Schleiden	(iii)	Discovered the nucleus
(D)	Schwann	(iv)	All plants are composed of different kinds of cell

(1) A-(i), B-(iii), C-(iv), D-(ii)
(2) A-(i), B-(iii), C-(ii), D-(iv)
(3) A-(iii), B-(i), C-(iv), D-(ii)
(4) A-(i), B-(iv), C-(ii), D-(iii)

19) **Assertion :** Cell theory says that all living organisms are composed of cells and product of cells.
Reason : On the basis of the number of cells present in the living organism. it is divided into two groups prokaryotes and eukaryotes.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) Both **Assertion** and **Reason** are false.
(4) **Assertion** is true but **Reason** is false.

20) **Assertion :** The presence of cell wall is a unique character of the plant cell.
Reason : Cells divided and new cells are formed from pre-existing cells.

(1) Both A and R are true and R is correct explanation of A.
(2) Both A and R are true but R is not a correct explanation of A.
(3) A is true and R is false
(4) Both A and R are false

21) **Assertion (A) :** Ribosomes are non-membrane bound organelles found in all cells.
Reason (R) : The prokaryotic cells lack membrane bound organelles.

(1) Both Assertion & Reason are correct and reason is correct explanation for assertion.
(2) Both assertion & reason are correct but reason is not correct explanation for assertion
(3) Assertion is correct but reason is incorrect

(4) Both assertion and reason are incorrect

22) Assertion A: Reductionist Biology involves understanding the molecular basis of physiological processes like digestion, excretion, and memory.
Reason R: In Reductionist Biology, the concepts and techniques of physics and chemistry are applied to study biological processes.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

23)

Which one of the following organisms is not an example of eukaryotic cells

(1) Amoeba proteus

(2) Paramecium caudatum

(3) Escherichia coli

(4) Euglena viridis

24) Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
(a) Schleiden, (a German), examined a large number of plants and observed that all plants are composed of different kinds of cells.
(b) Schwann, (a British) studied different types of animal cells and reported that cells had a thin outer layer which is today known as plasma membrane.
(c) Rudolf Virchow said "Omnis cellula-e cellula"
(d) Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.
Options :

(1) Statements (a) and (b) only

(2) Statements (b) and (c) only

(3) Statements (c) and (d) only

(4) Statements (a), (b), (c), (d) all

25) Find out incorrect match between Column – I and Column – II.

	Column - I		**Column - II**
(1)	Rudolf Virchow	(i)	Omnis cellula e cellula
(2)	Antonie van Leeuwenhoek	(ii)	First saw and described a live cell
(3)	Robert Brown	(iii)	First saw dead cell

(4)	Matthias Jakob Schleiden	(iv)	German Botanist

(1) 1
(2) 2
(3) 3
(4) 4

26) Animal cell differ from plant cell in possessing :

(1) Golgi body
(2) Centrosome
(3) Vacuole
(4) Plastid

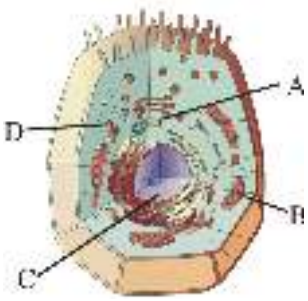
27) Outermost living boundary of animal cell is :-

(1) Cell wall
(2) Middle lamella
(3) Cell membrane
(4) Nuclear membrane

28) Membrane-bound organelles are absent in

(1) Saccharomyces
(2) Streptococcus
(3) Chlamydomonas
(4) Plasmodium

29) Identify the A, B, C and D in the given diagram.



	A	B	C	D
(1)	Centriole	Mitochondria	Nucleus	SER
(2)	Golgi apparatus	Centriole	Nuclear envelope	RER
(3)	Centriole	Mitochondria	Nuclear envelope	RER
(4)	Microtubules	Centriole	Nucleus	Golgi apparatus

(1) 1
(2) 2

(3) 3
(4) 4

30) What is the size of a typical eukaryotic cell :-

(1) 1–2 µm
(2) 10 – 20 µm
(3) 10 – 20 mm
(4) 1 – 2 mm

31)

The genetic material of the bacterial cell is localised within a discrete region, called as:-

(1) Nucleus
(2) Nucleolus
(3) Plasmid
(4) Nucleoid

32) Animal cell has a thin outer layer which is now known as "Plasma membrane". Who reported this?

(1) Schleiden
(2) Schwann
(3) Both (1) & (2)
(4) Robert Hooke

33) Which one of the following associated with prokaryotic cell ?

(1) Ribosomes
(2) Nucleus
(3) Chloroplast
(4) Lysosomes

34) Nucleoid is found in :-

(1) Blue green algae
(2) Algae
(3) Fungi
(4) Amoeba

35) Who proposed the theory that "cells arise only from pre-existing cells" ?

(1) Mohl
(2) Virchow
(3) Haeckel

(4) Brown

36) A plant cell has :–

(1) a large central vacuole and rigid cell wall
(2) a centriole for cell division
(3) a centrosome inactive in non-dividing cells
(4) absence of cell membrane

37) No organelles, like the ones in eukaryotes, are found in Prokaryotic cells except for -

(1) Membrane bound nucleus
(2) Inclusion bodies
(3) Gas vacuoles
(4) Ribosomes

38) Cell theory is not applicable for :-

(1) Viruses
(2) Animal
(3) plant
(4) Algae

39) Which of the following is seen only in prokaryotic cell?

(1) Lysosome
(2) Ribosome
(3) Mesosome
(4) ER

40) Result of Compartmentlization in cytoplasm of Eukaryotic cell is :-

(1) Cell Organelles
(2) DNA
(3) Cell Wall
(4) Cell Membrane

41)

Which structures perform the function of mitochondria in bacteria?

(1) Nucleoid
(2) Ribosomes
(3) Cell wall
(4) Mesosomes

42)

Prokaryotic cell does not have :-

(1) Nucleolus

(2) Membrane bound organelles

(3) Centriole

(4) All of these

43) Plant Cell and animal cells are different in how many given structures?
(A) Cell Wall
(B) Plastids
(C) Large Central Vacuole
(D) Centrosome (E) Food Storage

(1) 2

(2) 3

(3) 4

(4) 5

44) Larger subunit of prokaryotic ribosome is :-

(1) 30 S

(2) 40 S

(3) 50 S

(4) 60 S

45) Which of the following is not the character of prokaryotic cell ?

(1) Absence of membrane bound cell organelles

(2) Extensive compartmentalisation of cytoplasm

(3) Presence of 70s type of Ribosomes

(4) Presence of circular, ds DNA

46)

Consider the following four statements (A-D) and select the option which include all **correct** ones only :-
A. Compound epithelium is made up of one layer of cell and has role in secretion, absorption.
B. Simple cuboidal epithelium commonly found in ducts of glands and tubular part of nephron
C. Main function of simple squamous epithelium is forming a diffusion boundary.

D. Epithelial cells are loosely packed with intercellular matrix.

(1) A, B, C & D

(2) A & B

(3) B & C

(4) A, C & D

47) Tendon is an example of :-

(1) Dense regular connective tissue connecting muscle to skin
(2) Dense regular connective tissue connecting bone to bone
(3) Dense regular connective tissue connecting muscle to bone
(4) Dense irregular connective tissue connecting muscle to bone

48) Which of the following is incorrect for epithelium ?

(1) Cells are compactly packed with little intercellular matrix
(2) It is highly vascularised
(3) It provides a covering or a lining for some parts of the body
(4) It helps in secretion and absorption

49) "Germinativum layer" is known as:-

(1) Simple cuboidal epithelium which found in lining of ovary
(2) Inner most layer of stratified squamous epithelium found in epidermis of skin
(3) Outer most layer of stratified squamous epithelium found in buccal cavity
(4) Middle layer of transitional epithelium found in urinary bladder

50) Dry surface of skin, buccal cavity, Pharynx, Inner lining of duct of salivary gland & pancreatic duct is lined by :-

(1) Simple Epithelium
(2) Compound Epithelium
(3) Pseudostratified Epithelium
(4) Ciliated Epithelium

51) PCT of nephron is lined by :-

(1) Pseudostratified epithelium
(2) Stratified cuboidal epithelium
(3) Brush border cuboidal epithelium
(4) Non ciliated stratified columnar epithelium

52) Find out the correct match of column-I with column-II

	Column-I	**Column-II**
A	Stereociliated epithelium	Epididymis & Fallopian tube
B	Stratified non keratinized epithelium	Oesophagus & pharynx
C	Simple cuboidal epithelium	Pancreatic acini & Germinal epithelium

D	Neurosensory epithelium	Myoepithelium & cornea
E	Simple Columnar epithelium	Stomach

(1) A, B, E

(2) B C, E

(3) B, D, E

(4) A, C, F

53)

Which cells of epithelium get specialised for secretion in glandular epithelium ?
a - columnar cells
b - cuboidal cells
c - squamous cells

(1) Only a

(2) a and c

(3) b and c

(4) a and b

54) Given below are four statements (i) to (iv) each has two blanks. Select the option which correctly fills up the blanks in any two of these statements.
(i) The columnar epithelium is composed of ....(A).... and slender cells. Their ....(B).... are located at the base
(ii) Collagen fibres provide....(C)....and elastin fibres provide....(D)....and elasticity to the tissue
(iii) Adipose tissue is a....(E)....type of connective tissue located mainly beneath....(F)....
(iv) Tendon attach....(G)....to bones and ligament attach....(H)....to bones.

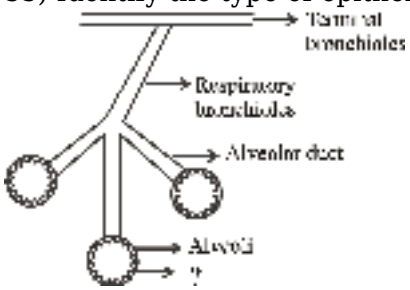
(1) (A) tall, (B) nucleus, (G) bones, (H) muscles

(2) (A) short, (B) organelles, (C) strength, (D) flexibility

(3) (C) strength, (D) flexibility, (E) loose, (F) skin

(4) (E) dense, (F) muscles, (G) bones, (H) muscles

55) Identify the type of epithelium tissue and their function :-



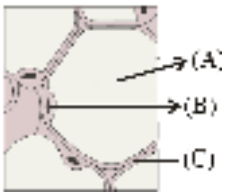
(1) Simple cuboidal epithelium = absorption

(2) Simple columnar epithelium = secretion

(3) Simple squamous epithelium = diffusion

(4) Pseudo stratified epithelium = Filtration

56) Identify A, B and C in the given diagram of adipose tissue. **[NCERT Pg : 103]**



(1) A-Cytoplasm, B-Nucleus, C-cellwall

(2) A-Fat storage area, B-mast cell, C-plasma membrane

(3) A-cell fluid, B-collagen fibres, C-plasmalemma

(4) A-Fat storage area, B-nucleus, C-plasma membrane

57) Our body is composed of almost four types of tissues. How many tissues in following are originated from mesodermal germ layers :-

(1) 3

(2) 2

(3) 1

(4) 4

58) Which of the following structures are lined by "Transitional epithelium' ?

(1) Alveoli

(2) Uterus

(3) Renal pelvis

(4) Stomach

59) Select the correct option about given diagram:-



(1) B = columnar epithelium - for protection and cover the dry surface of the skin

(2) B = squamous epithelium - found in tubular part of nephron

(3) A = squamous epithelium - found in wall of blood vessels

(4) A = cuboidal epithelium - found in air sac of lungs

60) Filteration and diffusion tissue is -

(1) Simple squamous epithelium

(2) Neuro-sensony epithelium

(3) Urothelium

(4) Pseudostratified epithelium

61)

Which of the following junction helps to stop substances from leaking across a tissue ?

(1) Interdigitation

(2) Gap junction

(3) Tight junction

(4) Desmosomes

62) The sketch given below represent a type of animal tissue. 

Which of the following statement is not related with above tissue ?

(1) It has a limited role in secretion and absorption

(2) It's main function is to provide protection against chemical and mechanical stresses.

(3) Cells are compactly packed with little intercellular matrix.

(4) It is found in the lining of stomach and intestine.

63) Our heart consists of :-
(i) Connective tissue
(ii) Epithelial tissue
(iii) Muscular tissue
(iv) Neural tissue

(1) Only (i)

(2) (ii) & (iii) only

(3) (i), (iii) & (iv) only

(4) All of these

64) Find out incorrect match :

	Tissue	Origin	Specific function
(1)	Epithelial	Ectodermal	Protection
(2)	Connective	Mesodermal	Connection
(3)	Muscular	Mesodermal	Compatibility
(4)	Nervous	Ectodermal	Irritability

(1) 1

(2) 2

(3) 3

(4) 4

65) Match column I and II and choose correct option :-

Column I		**Column II**	
A	Adhering junction	I	Help to stop substances from leaking across tissue

B	Gap junctions	II	Perform cementing to keep neighbouring cells together
C	Tight junctions	III	Facilitate the cells to communicate each other

(1) A-III, B-II, C-I

(2) A-II, B-III, C-I

(3) A-II, B-I, C-III

(4) A-I, B-III, C-II

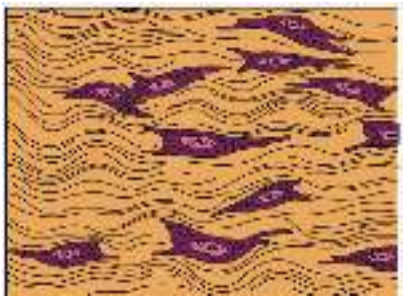
66) Tissue which present beneath the skin are ?

(1) Dense regular connective tissue

(2) Areolar tissue

(3) Adipose tissue

(4) Both (2) and (3)

67) 

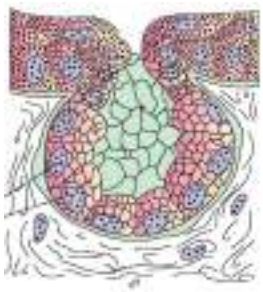
The given figure represent :-

(1) Adipose tissue

(2) Areolar tissue

(3) Cartilage

(4) Ligament

68) 

Identify the correct option regarding given figure :-

(1) Unicellular, Goblet cell

(2) Unicellular, Salivary gland

(3) Multicellular, Goblet cell

(4) Multicellular, Salivary gland

69) Read the following statements (A-D) :-

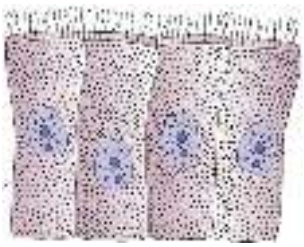
(A) The intercellular material of cartilage is solid and non-pliable and resists compression.

(B) Chondrocytes are enclosed in small cavities within the matrix.
(C) Most of the cartilages in vertebrate embryos are replaced by bones in adults.
(D) Bones have a hard and pliable ground substance rich in calcium salts and collagen fibres.
How many of the above statements are not incorrect ?

(1) One
(2) Two
(3) Three
(4) Four

70) **Statement-I :** Epithelium tissue has a free surface, which faces either a body fluid or the outside envinonment and thus provides a covering or lining for some part of the body.
Statement-II : In epithelium tissue, the cells are loosely packed with little intercellular matrix.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

71) 
The epithelium shown in above figure is present in :-

(1) Bronchioles
(2) PCT of nephron
(3) Air sacs of lungs
(4) All of these

72) Given below are two statement :- One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : In nearly all animal tissue, specialised junctions provide both structural and functional links between individual cells.
Reason (R) : Gap junctions help to stop substances from leaking across a tissue.
In the light of the above statements, choose the correct answer from the option given below:-

(1) (A) is not correct but (R) is correct
(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(4) (A) is correct but (R) is not correct.

73) Fill in the blanks and choose the correct option
(A) The cuboidal epithelium is composed of a single layer of ___(A)___ cells.
(B) Unicellular glands are consisting of ___(B)___ glandular cells.

(C) ___(C)___ epithelium provides protection against ___(D)___ and ___(E)___ stresses.

	(A)	(B)	(C)	(D)	(E)
(!)	Flat	Isolated	Compound	Mechanical	Immunological
(2)	Cube-like	Isolated	Compound	Mechanical	Chemical
(3)	Flat	Cluster of	Compound	Mechanical	Chemical
(4)	Cube-like	Cluster of	Simple	Immunological	Chemical

(1) 1
(2) 2
(3) 3
(4) 4

74)

Elastic cartilage is found in all except

(1) Epiglottis
(2) Ear pinna
(3) Nasal septum
(4) Eustachian tube

75)

Which of the following labelled structure represents macrophage in the given diagram?



(1) A
(2) B
(3) C
(4) D

76) **Assertion :** Epithelium faces external environment.
Reason : Epithelium forms covering of underlying tissue and has free surface.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

77) Structure present in outer ear joints is :-

(1) Cartilage
(2) Bone
(3) Cardiac muscle
(4) Osteocytes

78) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): Bones have a hard and non-pliable ground substance.
Reason (R): Matrix of bones is rich in calcium salts and collagen fibres

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is incorrect
(4) Both (A) and (R) are incorrect

79) Match the column-I with column-II and choose correct option :-

	Column-I		**Column-II**
(A)	Adipose tissue	(i)	Perform cementing to keep neighbouring cell together
(B)	Dense regular connective tissue	(ii)	Have a hard and non-pliable ground substance rich in calcium salts and collagen fibres
(C)	Bone	(iii)	Located mainly beneath the skin
(D)	Adhering junction	(iv)	Tendon and ligaments

(1) A-(iii), B-(iv), C-(i), D-(ii)
(2) A-(iv), B-(i), C-(iii), D-(ii)
(3) A-(iii), B-(ii), C-(i), D-(iv)
(4) A-(iii), B-(iv), C-(ii), D-(i)

80) Choose the correctly matched pair :-

(1) Inner lining of salivary duct - ciliated epithelium
(2) Moist surface of buccal cavity - glandular epithelium
(3) Tubular part of nephrons – cuboidal epithelium
(4) Inner surface of bronchioles – squamous epithelium

81) **Assertion:** Connective tissue is most abundant tissue in body.
Reason: Connective tissue help in secretion and absorption.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

82) A type of epithelial tissue which has thin flat cells, arranged edge to edge so as to appear like closely packed tiles, is found to be present at:

(1) Outer surface of ovary
(2) Inner lining of fallopian tube
(3) Inner lining of stomach
(4) Alveoli of lungs

83) Inflammatory substance produced during allergic reactions by mast cells is

(1) Histamine
(2) Heparin
(3) Serotonin
(4) Fibrinogen

84) **A:** Cell junctions may also be found in tissues other than epithelium.
B: Cells of cartilage are enclosed in small cavities.

(1) Both 'A' and 'B' are true
(2) A is true but B is false
(3) A is false but B is true
(4) 'A' and 'B' both are false

85) What will happen if ligaments are torn?

(1) Bones will move freely at joint and no pain
(2) Bone less movable at joint & pain
(3) Bone will become unfixed
(4) Bone will become fixed

86) Ligament is a

(1) Inelastic white fibrous tissue
(2) Modified white fibrous tissue
(3) Modified yellow elastic fibrous tissue
(4) None of these

87) **Assertion :-** Maximum reabsorption takes place in PCT.
Reason :- The epithelium of PCT of nephron has microvilli.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

88) **Statement-1 :** Connective tissues are most abundant and widely distributed in the body of complex animals.

Statement-2 : The matrix of connective tissue is made by modified polysaccharide.

(1) Only statement-1 is correct

(2) Only statement-2 is correct

(3) Statement-1 & 2 both are correct

(4) Statement-1 & 2 both are incorrect

89) Match the columns I, II and III and choose the correct combination from the options given.

	Column-I		**Column-II**		**Column-III**
a.	Tall and slender cells	1.	Cuboidal epithelium	K.	Bronchioles
b.	Flattened cells with irregular boundaries	2.	Columnar epithelium	L.	Tubular parts of nephrons
c.	Cilia on their free surface	3.	Squamous epithelium	M.	Lining of intestine
d.	Cube like cells	4.	Ciliated epithelium	N.	Walls of blood vessels

(1) a—2—M, b—3—N, c—4—K, d—1—L

(2) a—2—K, b—1—L, c—4—M, d—3—N

(3) a—4—M, b—3—L, c—1—K, d—1—N

(4) a—2—M, b—3—N, c—4—L, d—1—K

90) Inter vertebral disc are composed of :-

(1) Hyaline cartilage

(2) Elastic cartilage

(3) White fibrous cartilage

(4) Callified cartilage

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	1	3	3	1	2	4	2	1	1	3	2	4	2	2	2	2	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	3	2	2	2	2	1	3	1	3	3	2	4	3	1	1	1	3	2
Q.	41	42	43	44	45															
A.	2	4	3	3	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	3	3	1	4	3	3	4	1	2	4	3	3	2	3	2	1	1	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	1	2	1	2	3	4	3	2	2	1	4	2	2	2	3	1	3	3
Q.	86	87	88	89	90															
A.	3	2	4	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	1	2	1	1	4	2	4	3	4	1	2	1	2	2	1	1	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	4	3	2	3	2	1	2	4	2	1	1	2	1	1	1	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	4	3	2	3	3	2	2	2	3	2	4	3	3	4	1	3	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	4	3	2	4	4	4	2	2	1	4	2	3	4	1	1	1	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	4	1	1	3	3	1	3	1	3										

SOLUTIONS

PHYSICS

1) Enclosed area : $A = \pi r^2$

so $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

Here r = 8 cm, $\frac{dr}{dt}$ = 5 cm/s

$\Rightarrow \frac{dA}{dt} = (2\pi)\,(8)\,(5) = 80\pi \text{ cm}^2/\text{s}$

2)

Slope is zero at t = 0 & t = 2
Double derivative of y is –ve
at t = 2
Hence y attains maxima at t = 2

3) $\sin(75°) = \sin(45° + 30°)$
$= \sin45° \cos30° + \cos45° \sin30°$
$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \frac{1}{2}$
$= \frac{\sqrt{3}+1}{2\sqrt{2}}$

4) $3\int \cos x dx - 4\int \sin x dx$
= 3(sinx) – 4(– cosx) + C
= 3 sinx + 4 cosx + C

5) $y = \frac{x^3}{3}$
$\frac{dy}{dx} = \frac{3x^2}{3}$
$\frac{dy}{dx} = x^2$ (Parabola)

6) $S_\infty = \frac{a}{1-r}$
$a = 1, r = \frac{-1}{3}$
$S_\infty = \frac{1}{1-\left(-\frac{1}{3}\right)} = \frac{1}{1+\frac{1}{3}} = \frac{3}{4}$

7)

$y = 2x - x^2$

$\frac{dy}{dx} = 2 - 2x$

at x = 0 y = 0 and $\frac{dy}{dx} = 2$

as x ↑ $\frac{dy}{dx} \downarrow$

Curve will pass origin with positive slope but slope will decrease with increase in x.

8) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\frac{d}{dt}(t^2)}{\frac{d(5t)}{dt}} = \frac{2t}{5}$

9)

If $z = x + e \Rightarrow dz = dx$

$\int_{4e} \frac{dx}{x+e} = \int \frac{dz}{z} = \ell nz$

$\int_{3e} \frac{dx}{x+e} = \ell n(4e+e) - \ell n(3e+e)$

$= \ell n\left(\frac{5e}{4e}\right) = \ell n\left(\frac{5}{4}\right)$

10)

$x = R\cos\theta$

$y = R\sin\theta$

11)

$5x - 2y + 4 = 0$

$\Rightarrow 2y = 5x + 4$

$\Rightarrow y = \frac{5}{2}x + 2$ and c = +2

So, according to option, ans is option (1)

12)

$-1 \le \sin\theta \le 1$

$(\sin\theta)_{min} = -1$

$(\sin\theta)_{max} = +1$

So $(5 - 3\sin\theta)_{max} = 5 - 3(-1) = 8$

$(5 - 3\sin\theta)_{min} = 5 - 3(+1) = 2$

13) $\frac{d}{dx}\cos(2\sqrt{x}) = -\sin(2\sqrt{x}).\left(2\frac{1}{2\sqrt{x}}\right)$

$\begin{cases}\sqrt{x} = x^{+1/2} \\ \frac{d}{dx}\sqrt{x} = +\frac{1}{2}x^{+\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}\end{cases}$

$\frac{d}{dx}\cos(2\sqrt{x}) = \frac{-\sin(2\sqrt{x})}{\sqrt{x}}$

14) $y = -x^2 - 7.$

$\Rightarrow (y + 7) = -x^2$

15) $\bar{y} = \int_0^6 \frac{ydx}{6} = \frac{25}{3}$

16)

Angle rotate in 60 × 60 sec = 360 × 2π radian

Angle rotate in one second = $\frac{360 \times 2\pi}{60 \times 60}\frac{rad}{sec}$

17) $\cos\theta = \frac{2^2 + 5^2 - 4^2}{2(2)(5)} = \frac{13}{20}$

18) $\cos\left(\frac{11\pi}{6}\right) = \cos(330^\circ)$

$\Rightarrow \cos(300^\circ - 30^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$

19) Here, a = 4, d = 3

Last term = 94

a + (n – 1)d = 94

4 + (n – 1)3 = 94

(n – 1)3 = 90

n – 1 = 30

n = 31

Sum = $\frac{n}{2}$[First term + last term]

= $\frac{31}{2}$ [4+94] = 1519

20)

$12y = x^3$

$\Rightarrow 12\frac{dy}{dt} = 3x^2\frac{dx}{dt}$

$\Rightarrow \frac{dy}{dt} = \frac{x^2}{4} \cdot \frac{dx}{dt}$

When $\frac{x^2}{4} > 1$, then $\frac{dy}{dt} > \frac{dx}{dt}$

means y changes at a faster rate, when $\frac{x^2}{4} > 1$

$\Rightarrow x^2 > 4 \Rightarrow x > 2$

21)

$\log_{10}(32) = \log_{10}2^5$

$= 5\log_{10}2$

$= 5 \times 0.3010$

$= 1.505$

22)

$\sin\theta = \frac{1}{3} = \frac{P}{H}$

$\tan\theta = \frac{P}{B} = \frac{P}{\sqrt{H^2 - P^2}}$

$= \frac{P}{\sqrt{3^2 - 1^2}} = \frac{1}{2\sqrt{2}}$

23)

$(1 + x)^n \approx 1 + n\,x$

only if $x << 1$

24) $\vec{v} = |\vec{v}|\,\hat{v}$

$= 6\left(\frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}\right)$

$= 4\hat{i} + 4\hat{j} - 2\hat{k}$

25)

Check each option

$\left|\cos\theta\hat{i} - \sin\theta\hat{j}\right| = \sqrt{\cos^2\theta + \sin^2\theta}$

26) $\left|\overrightarrow{\Delta P}\right| = \left|-\vec{P} - \vec{P}\right| = \left|-2\vec{P}\right| = 2P$

27) $\vec{A} + \vec{B} = 3\hat{i} + 6\hat{j} - 2\hat{k}$

Unit Vector $\vec{A}+\vec{B}=\dfrac{\vec{A}+\vec{B}}{\left|\vec{A}+\vec{B}\right|}=\dfrac{3\hat{i}+6\hat{j}-2\hat{k}}{\sqrt{(3)^2+(6)^2+(-2)^2}}$

28) $\vec{AB}+\vec{AC}+\vec{AD}+\vec{AE}+\vec{AF}$

$=\vec{ED}+\vec{AE}+\vec{AC}+\vec{CD}+\vec{AD}\left[\begin{matrix}\vec{AB}+\vec{ED}\\ \vec{AF}+\vec{CD}\end{matrix}\right]$

$\vec{AD}+\vec{AD}+\vec{AD}=3\vec{AD}$

29) $\sqrt{A^2+B^2+2AB\cos\theta}=\dfrac{1}{2}\times\sqrt{A^2+B^2-AB\cos\theta}$

$4(A^2+B^2+2AB\cos\theta)=A^2+B^2-2AB\cos\theta$

$3A^2+3B^2+10AB\cos\theta=0$

$12B^2+3B^2+20B^2\cos\theta=0$

$20B^2\cos\theta=-15B^2$

$\cos\theta=-\dfrac{3}{4}$

$\theta=\cos^{-1}\left(-\dfrac{3}{4}\right)$

30)

$\cos\theta=-\dfrac{b}{a}$

$2R=S$

$4(a^2+b^2+2ab\cos\theta)=(a^2+b^2-2ab\cos\theta)$

$\dfrac{a}{b}=\sqrt{\dfrac{3}{7}}$

31)

as $\theta=60^\circ$

$=(\vec{a_1}-\vec{a_2})\cdot(2\vec{a_1}+\vec{a_2})$

$=2a_1^2-a_2^2-\vec{a_1}.\vec{a_2}$

$=2-1-\cos\theta$

$=\dfrac{1}{2}$

32)

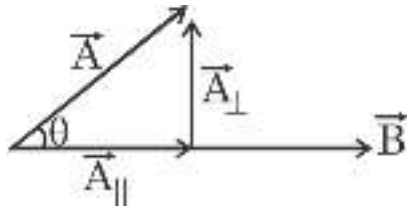
$\cos\theta=-\dfrac{A}{2B}$

$R^2=A^2+B^2+2AB\cos\theta$

$R=B$

33) $\frac{P+Q}{P-Q}=\frac{2}{1}$

34) $\hat{i},\hat{j}.\delta\hat{k}$ are not coplanar
all vector can not be coplanar



35)
$\hat{B}=\frac{3}{5}\hat{i}+\frac{4}{5}\hat{j}$

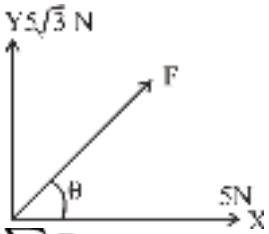
$\vec{A}_{||}=(A\cos\theta)\hat{B}=(\vec{A}\cdot\hat{B})\hat{B}=\frac{21}{25}\hat{i}+\frac{28}{25}\hat{j}\ \vec{A}_{\perp}=\vec{A}-\vec{A}_{||}=(\hat{i}+\hat{j}+5\hat{k})-\left(\frac{21}{25}\hat{i}+\frac{28}{25}\hat{j}\right)\frac{4}{25}\hat{i}-\frac{3}{25}\hat{j}+5\hat{k}$

36)

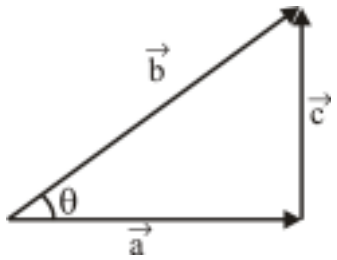
Let A and B be two forces
Greatest resultant = A + B = 10 .....(1)
Least resultant = A – B = 6 .... (2)
Solving (1) & (2) we get A = 8N and B = 2N when each force is increased by 3N
A' = A + 3 = 8 + 3 = 11 N
B' = B + 3 = 2 + 3 = 5 N

As the forces are acting at an angle of 90° So R' = $\sqrt{A'^2+B'^2}=\sqrt{(11)^2+(5)^2}=\sqrt{146}\,N$

37) $\sum F_x$ = sum of x-component of forces
= $2+\sqrt{3}\cos 30º+5\cos 60º-2\sin 30º$
= $2+\frac{3}{2}+\frac{5}{2}-1=5N$



$\sum F_y$ = sum of y-component of forces
= $\sqrt{3}\sin 30º+5\sin 60º+\sqrt{3}+2\cos 30º$
= $\frac{\sqrt{3}}{2}+\frac{5\sqrt{3}}{2}+\sqrt{3}+\sqrt{3}=5\sqrt{3}N$
$\therefore F=\sqrt{\left(\sum F_x\right)^2+\left(\sum F_y\right)^2}=\sqrt{5^2+(5\sqrt{3})^2}$
$=\sqrt{100}=10N$
$\therefore\tan\theta=\frac{\sum F_y}{\sum F_x}=\frac{5\sqrt{3}}{5}=\sqrt{3}\Rightarrow\theta=60$



38)

$\vec{a} + \vec{c} = \vec{b}$

$\cos\theta = \frac{\sqrt{3}c}{2c} = \frac{\sqrt{3}}{2}$ θ = 30° (Angle betwen $\vec{a}$ & $\vec{b}$) Angle between $\vec{a}$ & $\vec{b}$ = 60° and angle between $\vec{a}$ & $\vec{c}$ = 90°

39) $\hat{n} = \pm\frac{\vec{A} \times \vec{B}}{\left|\vec{A} \times \vec{B}\right|}$

40)

Area=$\left|\vec{A} \times \vec{B}\right|$

41)

Scalar component of $\vec{A}$ along $\vec{B}$,

$\vec{A} \cdot \hat{B} = (2\hat{i} + 3\hat{j}) \cdot \frac{(\hat{i} + \hat{j})}{\sqrt{2}}$

$= \frac{2+3}{\sqrt{2}} = \frac{5}{\sqrt{2}}$

42) $\vec{A}.\vec{B} = 0$

$\cos\left(\omega t - \frac{\omega t}{2}\right) = 0$

$t = \frac{\pi}{\omega}$

43)

AB sinθ = $\sqrt{3}$ AB cosθ ⇒ tanθ = $\sqrt{3}$

⇒ θ = 60°

$|\vec{A} + 2\vec{B}| = \sqrt{A^2 + (2B)^2 + 2(A)(2B)\cos 60^\circ}$

$= (A^2 + 4B^2 + 2AB)^{1/2}$

44) $\vec{A} \times \vec{B} = \vec{R}$

$\vec{R} \perp \vec{A}$

$\vec{R} \perp \vec{B}$

$\vec{R} \perp \left(\vec{A} + \vec{B}\right)$

So $\left(\hat{a}+\hat{b}\right)\cdot\left(\hat{a}\times\hat{b}\right)=0$

$=\left(\hat{a}+\hat{b}\right)\cdot\left(\hat{a}-\hat{b}\right)$

$=\hat{a}\cdot\hat{a}-\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{a}-\hat{b}\cdot\hat{b}$

$=1-1$

$=0$

45) As three vectors are coplanar, so $\begin{vmatrix} x & -2 & 3 \\ -2 & y & -4 \\ 0 & -z & 2 \end{vmatrix}=0$

$\Rightarrow x(2y-4z)-2(0+4)+3(2z-0)=0$

$\Rightarrow 2xy-4xz-8+6z=0$

$\Rightarrow xy-2xz+3z-4=0$

CHEMISTRY

46) Number of protons = $\left(\frac{10}{100}\right)$ (mole) $\times N_A \times 50$

$\Rightarrow \ 5\times 6.02\times 10^{23}$

$=3.01\times 10^{24}$

 number of proton in one molecule of $CaCO_3 = 20+6+24=50$

47)

Moles of Mg = $\frac{6}{24}=0.25$

0.25 moles of Na = $0.25\times 23=5.75$ g

48)

(i) m = 1g

(ii) $m = 18\times 1 = 18g$

(iii) $m = 23\times 1 = 23g$

(iv) $m=98$ amu $= 98\times 1.6\times 10^{-24}$ gram

49) $\frac{x}{30}\Rightarrow y\frac{2x}{60}\Rightarrow \frac{x}{30}\Rightarrow y$

50)

$n_{carbon} = n_{Ribose}\times 5$

$n_{Ribose} = \frac{1.25 \times 10^{24}}{6 \times 10^{23}} \times \frac{1}{5}$
=0.41
M=0.41 × 150 ~62 grm

51)

Question Explanation:
We're given the percentage of sulfur in an organic compound and asked to find the minimum possible molecular weight of that compound. The key here is understanding that molecular weights must be whole number multiples of atomic weights. Since sulphur's atomic weight is 32, the minimum molecular weight will be the one where the 4% sulfur corresponds to exactly one sulphur atom.

Given Data:

A. Percentage of sulfur = 4%

B. Atomic weight of sulfur = 32 (approximately)

Concept: The percentage of an element in a compound.

Mathematical Calculation:
Let M be the minimum molecular weight of the compound. We can set up the equation:

(Mass of sulphur/M) × 100 = Percentage of sulphur

(32/M) × 100 = 4

3200 = 4M

M = 3200/4

M = 800

Final Answer:
The minimum molecular weight of the organic compound is 800. Therefore, option 3 is correct.

52) mole = $\frac{\text{volume of gas}}{22.4} = \frac{2.24}{22.4}$ = 0.1 mole
$0.1 = \frac{4.4}{M}$; M = 44
So gas will be N_2O
14 × 2 + 16 = 44

53)

Moles of hydrogen atom = $4 \times {}^{n}C_2H_4O_2$
= 4 × 4 = 16 moles
mass = 16 × 1 = 16 g

54)

no of He atom = $\frac{16amu}{4amu}$

55)

Generated by Allie

Problem Statement: The question asks to find the empirical formula of an organic compound given the percentage composition of carbon (C), hydrogen (H), and oxygen (O). It requires converting the given mass percentages into the simplest whole number mole ratio for the elements.

Underlying Concept: The core concept is the empirical formula determination in chemistry, which involves converting percent composition into moles, then dividing by the smallest mole value to find the simplest whole number ratio representing the compound's formula.

Relevant Formulas: Moles of element (n) = Mass of element (g) / Atomic mass (gÂ·molâ»Â¹)

Step-by-Step Calculation: 1. Assume a sample size of 100 g to directly use percentage as mass: - C mass = 40.0 g - H mass = 6.7 g - O mass = 100 - (40.0 + 6.7) = 53.3 g

2. Calculate moles of each element:

$$n(C) = \frac{40.0}{12.0} = 3.33 \text{ mol} \quad n(H) = \frac{6.7}{1.0} = 6.70 \text{ mol}$$

$$n(O) = \frac{53.3}{16.0} = 3.33 \text{ mol}$$

3. Divide by smallest mole number (3.33):

$$C : H : O = 1 : \frac{6.70}{3.33} : 1 = 1 : 2 : 1$$

Thus, the empirical formula is CH_2O.

Tips and Tricks: Remember that the empirical formula shows the simplest whole number ratio of atoms, so always divide mole quantities by the smallest mole value.

Common Mistakes: A frequent error is neglecting to subtract the given percentages from 100% to find the oxygen percentage. Another is forgetting to divide all moles by the smallest mole value for simplicity.

Why Other Options Are Incorrect?: 'CH' does not account for oxygen present in the compound. 'C_3H_6O' and 'C_3H_4O_2' do not match the ratio derived from the given percentages, so they are incorrect empirical formulas for this composition.

56)

(i) $n_{Co_2} = \frac{22.4 \times 10^3}{22400} = 1$

(ii) $n_{Co_2} = \frac{22}{44} = \frac{1}{2}$

(iii) $n_{Co_2} = \frac{11.2}{22.4} = \frac{1}{2}$

(iv) $n_{Co_2} = 0.1$

57)

Generated by Allie

Problem Statement: The question is asking to find the molecular weight (molar mass) of a gas when its density at Standard Temperature and Pressure (STP) is given as 0.2 grams per liter (g/L).

Underlying Concept: The core concept involves the relationship between the density of a gas at STP and its molar mass. At STP, one mole of any ideal gas occupies a volume of 22.4 liters. Using this, the molar mass can be calculated from the density by multiplying the density by the molar volume.

Relevant Formulas: Molar mass (M) = Density at STP (d) Ã— Molar volume at STP (V) where, d is in g/L, V = 22.4 L/mol at STP, M will be in g/mol.

Step-by-Step Calculation: Given density d = 0.2 g/L At STP, molar volume V = 22.4 L/mol M = d Ã— V = 0.2 Ã— 22.4 = 4.48 g/mol

Tips and Tricks: Recall the molar volume of an ideal gas at STP as 22.4 L/mol. Multiplying density by 22.4 immediately gives molar mass without needing to convert units.

Common Mistakes: Students might forget that the molar volume at STP is 22.4 L, or confuse the units of density leading to incorrect calculations.

Why Other Options Are Incorrect?: Options like 2.24 or 10 g/mol do not match the correct calculation using the given density and molar volume. For example, 2.24 would arise if one mistakenly took half the molar volume or ignored units.

58)

Ans : 46

59)

Total volume of 10 drop = 0.0018 × 10 = 0.018 ml
Mass = 0.018 × 1 = 0.018 g

moles = $\frac{0.018}{18} = 10^{-3}$

60)

$n_{urea} = \frac{120}{60} = 2$

$V_{solution} = \frac{m}{d} = \frac{1120}{1.14} = 982.4ml$

Molarity = $\frac{2}{982.4} \times 1000 = 2.04$

61) Mol mass of $C_3XO_4 = \frac{10}{0.05} = 200$

$(12 \times 3) + M_X + (4 \times 16) = 200$

$M_x = 100$

62) Given : Percentage weight of nitrogen = 28 ; Percentage weight of metal = 72 ; No. of atoms of metal = 3 and number of atoms of nitrogen = 2.
We know that when 3 atoms of metal combine with 2 atoms of nitrogen, then molecular formula of the compound will be M_3N_2. Let x be the atomic weight of metal.
We also know that 28 g of nitrogen combines with 3x g of metal Or 28 g of nitrogen combines with 72 g of the metal. Therefore 3x = 72 or x = 24.

63)

Definition

64) $PV = nRT$

$\frac{152}{760} \times V = 0.1 \times 0.0821 \times 300$

$V = 12.31$ L

65) Mass of Au = $\frac{25}{10} \times \frac{40}{10} \times \frac{0.25}{10} \times 19.32$

moles of Au = $\frac{25 \times 40 \times 0.25 \times 19.32}{197 \times 1000}$

No. of Au atoms = $\frac{25 \times 40 \times 0.25 \times 19.32}{197 \times 1000} \times N_A$

66)

The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is 68.4 g.

67) Given : Weight of oxygen in first oxide = 27.6% ; Weight of oxygen in second oxide = 30.0% and formula of first oxide = M_3O_4.
We know that weight of metal in first oxide = 100 â€“ 27.6 = 72.4% and weight of metal in second oxide = 100 â€“ 30 70%
We also know that as the first oxide in M_3O_4, therefore no. of atoms of 72.4 weight of M = 3 and no. of atoms of 27.6 weight of O = 4.
Thus in second oxide, no of atoms of 70 weight of M = $\frac{3}{72.4}$ Ã— 70 = 2.9
Similarly, no. of atoms of 30 weight of $O = \frac{4}{27.6} \times 30 = 4.35$.
Since the ratio of M : O in the second oxide = 2.9 : 4.35 = 1 : 1.5 = 2 : 3, therefore the formula of second oxide is M_2O_3.

68)

C H O
2 6 15

$C_2H_6H_{15}$

69)

Mass loss on heating due to vaporization of water hence.

$$24.5 = \frac{18x}{111+18x} \times 1000$$

$x = 2$

70)

Let V mL of alcohol be requried
∵ mass of alcohol is same in both solutions

$$\therefore \quad \frac{75}{100} \times 0.8 \times V = \frac{30}{100} \times 0.9 \times 150$$

$$V = 67.5 \text{ mL}$$

71)

x = 40 g
y = 60 g
atomic mass x = 0.6a
atomic mass y = a

$$\text{Mole Ratio} = \frac{20}{6} : \frac{3}{1}$$

$x_{10\frac{1}{9}} = 10 : 9$

72)

Molarity = $\frac{\left(\frac{38}{36.5}\right)}{\frac{100}{1.19}} \times 1000$
=12.37 M

$1.5 = \frac{3x+y}{x+y}$ or $1.5x + 1.5y = 3x + y$

or $1.5x = 0.5y$ or $\frac{x}{y}=\frac{1}{3}$

73)

When the solution is mixed in the ratio $\frac{y}{x}$ i.e., $\frac{3}{1}$

$M = \frac{3\times3+1\times1}{4}=\frac{10}{4}= 2.5$

74)

Defination

75)

Molarity for H_2O_2 solution = $\frac{\text{Volume Strength}}{11.35}$
For mixture,
$M_f = \frac{M_1 \cdot V_1 + M_2V_2}{V_1 + V_2}$
$= \frac{\left(\frac{20}{11.35}\right) \times 50 + \left(\frac{10}{11.35}\right) \times 200}{50 + 200}$
$= \left(\frac{3000}{11.35}\right) \times \frac{1}{250}$
$= \left(\frac{12}{11.35}\right)$
Volume Strength = 11.35 × Molarity
$= 11.35 \times \frac{12}{11.35}$
= 12

76) $5 = \frac{15/40}{\frac{100}{d}} \times 1000$

d = 1.33

77)

Remaining mass of solute = 80 – 50 = 30
$= \frac{30}{350} \times 100$ = 8.57%

78)
$$W_{mix} = W_X + W_Y$$
$$= (n \times M)_X + (n \times M)_Y$$
$$3.72 = 0.02 \times M_X + (0.06 - 0.02) \times 48$$
$$M_X = 90$$

79)

Molecular Mass = $13 \times 12 + 12 \times 1 + 3 \times 16 + \frac{3.02 \times 10^{-23}}{1.66 \times 10^{-24}}$
=234 amu

80)

Mass of 10 moles of H_2O = 180
$$\text{Mole} = \frac{180}{20} = 9$$

81) $$VD = \frac{d_{gas}}{dH_2} = \frac{0.001293 g/ml}{\frac{2g}{22400}\ ml} = 14.3$$

82)

$$24.31 = \frac{79 \times 24 + x \times 25 + (21 - x) \times 26}{100}$$
x = 11
=26 mg = 21 – 11 = 10

83) $\overline{Mw} = 2 \times 34.5 = 69$
69 = x × 46 + (1-x) × 92
x = 0.5
50%

84) Given : Weight of the carbon monoxide (CO) = 2.8 g; pressure of the carbon monoxide = 0.821 atm; volume of carbon monoxids = 3 litres and gas constant (R) = 0.0821 L-atm-mol$^{â€“1}$-K$^{â€“1}$ we know that molecular weight of the carbon monoxide (CO) = 12 + 16 = 28.
Therefore no. of moles in 2.8 g of the carbon monoxide.
$$n = \frac{\text{Weight of carbon monoxide}}{\text{Molecular weight of carbon monoxide}}$$
$$= \frac{2.8}{28} = 0.1$$
We also know from the ideal gas equation that temperature of the carbon monoxide.
$$T = \frac{PV}{nR} = \frac{0.821 \times 3}{0.1 \times 0.0821} = 300\ K$$

85) $d = \frac{PM_w}{RT}$

$1.33 = \frac{(764/760)M_w}{0.0821 \times 423}$

$M_w = 46$ (NO_2)

86)

Mw = 2 × 70 = 140

140 = (12 + 15)x

x = 5

87)

under similar condition

$V \propto n$

$\frac{v_1}{v_2} = \frac{n_1}{n_2} = \frac{0.44}{0.32} = \frac{0.2/Mw}{\left(\frac{0.1}{44}\right)}$

Mw = 64

SO_2 gas

88)

Total moles = $\frac{16}{32} + \frac{3}{2}$

$= \frac{1}{2} + \frac{3}{2} = 2$

volume = 2 × 22.4 = 44.8 L

=44800 ml

89) PV = nRT

log P + log V = log (nRT)

log P = log (RT) – log V

90) $\overline{Mw} = \frac{16 \times 1 + 28 \times 1}{2}$

$= \frac{44}{2} = 22$

$d = \frac{22}{22.4} = 0.98 g/L$

BIOLOGY

91) NCERT-XI Pg. # 125,126

92) **Explanation:** Erythrocytes (red blood cells) of mammals do not have a nucleus, unlike other cell types.
The correct answer is:
(3). Erythrocyte of mammals

93)

NCERT-XI, Pg. # 98

94)

NCERT-XI, Pg. # 98

95)

Generated by Allie

Problem Statement: The question asks to identify which statements about cells and cell theory are correct and which are incorrect, and then select the matching choice that states how many statements are correct and how many are incorrect.

Underlying Concept: The core concept involved is the Cell Theory, which is a fundamental principle in biology stating that all living organisms are composed of cells, cells arise from pre-existing cells, and studying the contributions of scientists like Schwann and Virchow who helped develop this theory.

Tips and Tricks: Remember: Schwann studied animal cells, Schleiden studied plant cells, and Virchow coined 'omnis cellula-e cellula' meaning all cells arise from existing cells.

Common Mistakes: Confusing the professions or contributions of Schwann (zoologist, not botanist) and Schleiden (botanist). Misattributing the phrase 'omnis cellula-e cellula' to Schleiden instead of Virchow.

Why Other Options Are Incorrect?: Statement (B) is incorrect because Schwann was a British zoologist who studied animal cells, not a botanist examining plants. Statement (D) is incorrect because the phrase 'omnis cellula-e cellula' was coined by Rudolf Virchow, not Schleiden.

96)

The core concept involved is the Cell Theory, which is a fundamental principle in biology stating that all living organisms are composed of cells, cells arise from pre-existing cells, and studying the contributions of scientists like Schwann and Virchow who helped develop this theory.

97)

Cell is composed of small amount of cytoplasm with nucleus and surround by a plasma membrane with or without cell wall.

98)

Generated by Allie

Problem Statement: The question asks to identify the correct year and the scientist who examined many plants and observed that all plants are made up of different kinds of cells which form plant tissues.

Underlying Concept: This question relates to cell theory in biology, specifically the early discovery that plants are composed of cells. Matthias Schleiden, a German botanist, was one of the scientists who contributed to this idea by studying plant tissues and proposing that plants are made of cells.

Tips and Tricks: Remember that Schleiden is specifically known for plant cells, while Schwann contributed to animal cells; Leeuwenhoek worked much earlier and was not related to the plant tissue observation in this period.

Common Mistakes: Confusing Schwann with Schleiden; mixing up years (1838 vs. 1839); attributing the discovery to Leeuwenhoek who was from an earlier era.

Why Other Options Are Incorrect?: 1839, Leeuwenhoek is incorrect because Leeuwenhoek lived much earlier (17th century). 1839, Schwann is wrong because Schwann studied animal cells primarily. 1838, Schwann is also wrong for the same reason. The correct answer is 1838, Schleiden who studied plant cells.

99)

NCERT Page No. 126

100)

NCERT, XI Pg # 87

101)

NCERT, XI Pg # 87

102)

NCERT XI Pg . # 88

103)

NCERT, XI Pg # 88

104)

Cytoplasm is main arena for cellular activities.

105) NCERT Pg # 123E

106)

Cell is composed of small amount of cytoplasm with nucleus and surround by a plasma membrane with or without cell wall.

107) **[NCERT NO. 88]**

108)

NCERT (XI) Pg. # 125-126

109)

NCERT XI Pg. # 125, 126

110) XI NCERT Pg. No. # 126

111) NCERT, Pg. # 126

112) The assertion is true as Reductionist Biology does focus on understanding the molecular basis of physiological processes. The reason is also true and is the correct explanation for the assertion because Reductionist Biology indeed applies the concepts and techniques of physics and chemistry to understand biological processes, which is the essence of the physico-chemical approach mentioned in the text.

113)

114)

Schwann studied animal cells, Schleiden studied plant cells, and Virchow coined 'omnis cellula-e cellula' meaning all cells arise from existing cells.

115)

Schwann studied animal cells, Schleiden studied plant cells, and Virchow coined 'omnis cellula-e cellula' meaning all cells arise from existing cells.

116)

Answer: 2. Centrosome

Explanation:
Animal cells differ from plant cells in possessing a centrosome, which is involved in organizing microtubules and is crucial for cell division. Plant cells typically lack centrosomes and instead have structures like plastids and large vacuoles.

117)

NCERT XI Pg # 88

118)

Streptoccus is a prokaryote

119) NCERT XI Pg. # 130

120)

NCERT (XIth) Pg. # 127

121)

Nucleoid is false nucleus that lack nucleas memberane.

122) NCERT XI Pg. # 126, Para 1 line 1,2

123)

NCERT-XI, Pg. # 91

124) **Key Analysis:**

A. Nucleoid is a region in prokaryotic cells where the genetic material (DNA) is located. It is not enclosed by a membrane, unlike the nucleus in eukaryotic cells.

1. Blue-green algae (also known as cyanobacteria):
   A. These are prokaryotes, and they have a nucleoid instead of a nucleus.
2. Algae:
   A. Most algae are eukaryotes, so they have a true nucleus, not a nucleoid.
3. Fungi:
   A. Fungi are also eukaryotes and have a true nucleus.

4. Amoeba:

A. Amoeba is a eukaryotic organism with a true nucleus.

Correct Answer:
1. Blue-green algae

125) Module Pg. # 126

126)

The correct answer is : 1. a large central vacuole and rigid cell wall

Explanation: Plant cells have a large central vacuole for storage and a rigid cell wall for structure. Other options describe features not typical of plant cells.

127) NCERT (XIth) Pg. # 128

128) **The correct answer is: 1. Viruses Explanation:** Cell theory states that all living organisms are composed of cells, and all cell functions occur within cells. However, viruses do not follow this principle because they are not made of cells. They are without cell and cannot carry out life processes independently; they require a host cell for replication. Therefore, cell theory is not applicable to viruses.

129)

Mesosome is seen only in prokaryotic cell.

130)

NCERT-XI, Pg. # 91

131)

The structure that performs a similar function to mitochondria in eukaryotic cells is the mesosome

132)

Prokaryotic cells do not have nucleolus, centrioles, or membrane-bound organelles. Therefore, the answer is all of these

133)

Explanation: Plant and animal cells differ in 5 structures:
1. Cell Wall (plant cells only)
2. Plastids (plant cells only)

3. Large Central Vacuole (plant cells only)
4. Centrosome (animal cells only)
5. Food Storage (starch in plants, glycogen in animals)
The correct answer is (4) 5

134) **Explanation:** The larger subunit of a prokaryotic ribosome is the 50S subunit, while the smaller subunit is the 30S subunit. Together, they form the complete 70S ribosome in prokaryotes. **The correct answer is: (3)** 50 S

135) NCERT XI Pg.# 127-129

136) NCERT (XIth) Pg. # 101, 102

137) NCERT XI Pg # 103

138)

NCERT (XIth) Pg. # 101

139) Module-2 Pg. # 84

140) NCERT Pg. # 102

141) NCERT Pg. # 101, para 2

142) NCERT Pg# 101, 102, para-2

143)

NCERT (XIth) Pg. # 102

144) NCERT Pg# 101 and 103

145) NCERT Pg# 101

146) NCERT Pg# 103, fig.-7.4(b)

147) NCERT Pg. # 100

148) NCERT (XI), Pg. # 102

149)

NCERT Page # 101

150)

Simple squamous epithelium is the tissue primarily associated with filtration and diffusion.

151)

NCERT XI, Pg. No. 102

152)

NCERT-XI, Pg. # 102

153) NCERT-XI, Pg. # 106

154)

Let's analyze each match:

Muscular, Mesodermal, Compatibility: This is incorrect. Muscular tissue is indeed derived from the mesoderm, but its specific function is contraction and movement, not compatibility.

Therefore, the incorrect match is (3) Muscular, Mesodermal, Compatibility.

155) NCERT (XIth) Pg. # 102

156)

NCERT XI, Pg. # 103

157) NCERT (XI) Pg. # 103

158) NCERT Pg. # 102

159) NCERT Pg. # 104

160) NCERT Pg. # 101

161) NCERT Pg. # 101

162) NCERT-XII, Pg. # 102

163) NCERT PAGE (ENGLISH): 101,102

164)

NCERT Page # 104

165)

A-Mast cells, B-Collagen, C-Fibroblast, D- Macrophage

166)

The correct answer is option 1: Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

Explanation:

Assertion: "Epithelium faces the external environment of the body." → This is true because epithelial tissue lines the outer surfaces of organs and the body, as well as internal cavities.

Reason: "Epithelium forms a covering of underlying tissue and has a free surface." → This is also true because epithelial tissue covers body surfaces and organs, providing protection and allowing interactions with the external environment.
Since the reason correctly explains why the assertion is true, option 1 is the correct choice.

167)

Explanation: The outer ear (pinna) is primarily made up of elastic cartilage, which provides flexibility and shape. Cartilage is also found in joints to provide cushioning and reduce friction between bones.

168)

NCERT PAGE : 104

169) NCERT-XI, Pg # 102, 103, 104

170) Module

171) NCERT-XII, Pg. # 102

172) Inner lining of cheeks is stratified keratinised squamous epithelium. This epithelium is compound epithelium in which cells of superficial layer are squamous, i.e. polygonal and

flattened in outline.

173) Mast cell contain granules of heparin and histamine. Histamine produces inflammation increases permeability of blood vessels, activates blood platelets and causes contraction of smooth muscles. Heparin prevents blood clotting.

174) Fact

175) Ligaments are structure which connect bone to bone. They are made of bundles of elastic fibres and few collagen fibers

176) Ligaments are yellow elastic fibrous (dense regular) connective tissue. Ligaments are elastic in nature where matrix is pervaded by bundles of elastic fibres and collagen fibres.

177) NCERT-Page No. # 101

178) NCERT-XII, Pg. # 102, 103

179)

NCERT - XI. Pg # 101

180) **Concept :**
Work fiborous cartilage

Solution/Explanation :
Perichondrium is absent because complete WFCT is converted into cartilage. In matrix bundle of collagen fibres are more in quantity so it is strongest cartilage.
Intervertebral disc: A pad of cushion like structure which absorb mechanical shock & jerks and protect vertebral column. Central part of this disc is soft called as Nucleus pulposus